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SafeMyanmar

Project Category

Mobile And Ubiquitous Computing Project

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Sep 6, 2026	0.02	Project Group.	Added references and reviewed the report
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Keywords

SafeMyanmar, Mobile Computing, Ubiquitous Computing, Disaster Response, Earthquake Information, GPS, Context Awareness, Edge Computing, Cloud Computing, Artificial Intelligence, High Performance Computing, Emergency Assistance, SOS, Offline Computing, Flutter, FastAPI, PostgreSQL

Introduction

Background

Myanmar is highly vulnerable to natural disasters such as earthquakes, floods, cyclones, landslides, and fires. These events often occur unexpectedly, leaving citizens with limited access to trusted information, safe evacuation routes, or immediate assistance. Traditional communication channels, such as social media, are prone to misinformation, which can worsen panic and delay rescue operations.

Advances in Mobile Computing, Ubiquitous Computing, GPS, Edge Computing, Cloud Computing, Artificial Intelligence (AI), and High Performance Computing (HPC) provide opportunities to design smart disaster-response systems. SafeMyanmar leverages these technologies to deliver fast, reliable, and context-aware emergency services tailored to Myanmar's needs.

Problem Statement

Disaster response in Myanmar faces several critical challenges:

- **Lack of Trusted Information:** Citizens often rely on unverified social media posts, leading to misinformation.
- **Difficulty Finding Safe Routes:** Blocked roads, flooded areas, and fire zones hinder evacuation.
- **Slow Emergency Communication:** Injured or trapped individuals struggle to share their location.
- **Limited Access to Guidance:** Internet connectivity may fail, restricting access to online help.
- **Lack of Context-Aware Assistance:** General disaster advice does not adapt to the user's real-time location and situation.

These issues highlight the urgent need for a **smart, context-aware mobile application** that integrates real-time data, AI-powered guidance, and GPS navigation to reduce disaster response time and save lives.

Project Overview

SafeMyanmar is a **mobile disaster-response application** designed to assist users before, during, and after emergencies. It provides:

- **Trusted disaster alerts** from verified sources.
- **GPS-based safe route navigation** considering blocked roads and shelters.
- **SOS emergency button** for quick communication.
- **Rescue beacon mode** for trapped victims.
- **AI-powered emergency assistant** for first aid and disaster guidance.
- **Trusted disaster reports** from government and humanitarian organizations.

The system architecture integrates **cloud computing** for large-scale data management, **edge computing** for fast local processing, and **AI** for personalized emergency recommendations.

Objectives

The objectives of SafeMyanmar are:

- **Provide Trusted Alerts** from government and disaster agencies.
- **Enable Safe Navigation** using GPS and real-time updates.
- **Facilitate Emergency Assistance** through SOS communication.
- **Support Rescue Teams** with GPS-based victim location.
- **Deliver AI Guidance** for first aid and disaster safety.
- **Reduce Response Time** by minimizing delays in communication.
- **Demonstrate Computing Concepts** through real-world disaster applications.

Project Scope

SafeMyanmar covers multiple disaster scenarios and integrates diverse computing technologies.

Table 1.1: Scope of SafeMyanmar Features

Feature	Scope
Disaster Alerts	Verified alerts from trusted agencies
Safe Route Navigation	GPS-based evacuation routes with real-time updates
SOS Button	Emergency location sharing with contacts
Rescue Beacon	Alarm, flashlight SOS, HELP display
AI Assistant	First aid and disaster guidance
First Aid Guide	Offline access to CPR, burns, bleeding, fractures
Trusted Reports	Disaster reports from official organizations
Damage Reporting	User-submitted disaster damage reports (future)

Target Users

SafeMyanmar is designed for diverse user groups:

- **Citizens:** Receive alerts, safe routes, and first aid guidance.
- **Families:** Coordinate safety and track loved ones.
- **Students:** Learn disaster safety and first aid.
- **Rescue Teams:** Locate victims and coordinate rescue.
- **Volunteers:** Support disaster response efforts.
- **Government Agencies:** Manage disaster information and response.

System Analysis

Functional Requirements

The system must provide the following **functional capabilities**:

Table 2.1: Functional Requirements of SafeMyanmar

Requirement	Description
Disaster Alerts	Receive verified alerts from trusted sources.
Safe Route Navigation	Recommend evacuation routes based on GPS and disaster conditions.
SOS Communication	Send location and emergency messages to contacts.
Rescue Beacon	Activate alarm, flashlight SOS, and HELP display.
AI Assistant	Provide first aid and disaster guidance.
First Aid Guide	Offline access to CPR, burns, bleeding, fractures.
Trusted Reports	Display disaster reports from official agencies.

Non-functional Requirements

SafeMyanmar must also meet **non-functional requirements** to ensure reliability and usability:

Table 2.2: Non-functional Requirements of SafeMyanmar

Requirement	Description
Performance	Fast response times using edge computing.
Reliability	Continuous operation during disasters with offline support.
Scalability	Handle large-scale disaster data via cloud computing and HPC.
Security	Ensure data integrity and protect user privacy.
Usability	Simple interface for citizens, families, and rescue teams.
Availability	Operate even with limited network connectivity.

Main Features

SafeMyanmar integrates multiple emergency features into one application:

- **Trusted Disaster Alerts:** Verified alerts from government and disaster agencies.
- **GPS Safe Route Navigation:** Context-aware evacuation routes.
- **SOS Emergency Button:** Quick emergency communication.
- **Rescue Beacon Mode:** Alarm, flashlight SOS, HELP display.
- **AI Emergency Assistant:** First aid and disaster guidance.
- **First Aid Guide:** Offline access to essential medical instructions.
- **Trusted Disaster Reports:** Reliable disaster updates from official organizations.

System Architecture and Design

Overall Architecture

SafeMyanmar integrates multiple computing paradigms into a layered architecture:

Trusted Data Sources: Government agencies, weather services, earthquake APIs, disaster organizations.

Cloud Layer: Centralized disaster database, AI services, notification system, and user authentication.

Edge Layer: Fast local processing for safe route calculation and context-aware decision-making.

Mobile Application: User-facing features such as alerts, navigation, SOS, rescue beacon, AI assistant, and first aid guide.

Mobile Computing

SafeMyanmar demonstrates **mobile computing concepts** by leveraging smartphone hardware and connectivity:

- **GPS** for location detection.
- **Mobile Sensors** such as camera, flashlight, accelerometer.
- **Wireless Communication** via 4G/5G, Wi-Fi.
- **Push Notifications** for instant disaster alerts.
- **Mobility** ensuring continuous support while users move during emergencies.

Ubiquitous Computing

SafeMyanmar is a **context-aware application**, continuously adapting to the user's situation:

- Detects **user location**.
- Understands **current disaster situation**.
- Identifies **nearest safe shelter**.
- Provides **appropriate recommendations** automatically without manual search.

This ensures personalized and timely assistance during emergencies.

GPS and Location Services

GPS plays a central role in SafeMyanmar:

- **Real-time location detection.**
- **Safe route navigation** considering blocked roads, flooded areas, and fire zones.
- **Rescue team support** by sharing victim coordinates.

Edge Computing

Edge computing reduces reliance on distant servers by processing data locally:

- **Route calculation** near the user.
- **Disaster information processing** for faster response.
- **Reduced latency** during emergencies.

Cloud Computing

The cloud layer provides centralized services:

- **User authentication.**
- **Disaster information database.**
- **AI services** for advanced guidance.
- **Real-time synchronization** across devices.
- **Push notifications** for alerts.
- **Historical data storage** for analysis.

Artificial Intelligence

AI enhances SafeMyanmar by providing:

- **Emergency assistance** (e.g., earthquake safety, first aid).
- **Disaster safety recommendations.**
- **Safe route suggestions.**
- **Question answering** for user queries.

High Performance Computing

HPC is used on the cloud side to process large-scale disaster data:

- **Earthquake simulations.**
- **Flood prediction models.**
- **Evacuation route optimization.**
- **AI model training.**

HPC ensures faster, more accurate disaster forecasting and decision-making compared to traditional computing systems.

System Implementation

Technology Stack

Table 4.1: Technology Stack

Layer	Technologies	Purpose
Mobile Application	Flutter and Dart	Android interface and mobile functionality
State and Navigation	Riverpod and GoRouter	Application state and screen navigation
Backend	Python and FastAPI	REST API and data processing
Main Database	PostgreSQL and SQLAlchemy	Store normalized earthquake information
Local Storage	Drift and Flutter Secure Storage	Offline cache, profile and emergency contacts
Earthquake Source	USGS GeoJSON Feed	Retrieve earthquake observations
Maps and Routes	Mapbox	Display maps and request route alternatives
Communication	SMS and Bluetooth Low Energy	Emergency-message and nearby SOS sharing
AI Assistance	Offline knowledge base and optional on-device models	Emergency information and question classification

Mobile Application

The mobile application is the primary interface for users. It provides disaster alerts through push notifications, GPS-based safe route navigation, and emergency features such as the SOS button and Rescue Beacon Mode. The app also integrates the AI Emergency Assistant and an offline first aid guide, ensuring that users can access critical information even when connectivity is limited.

Backend

The SafeMyanmar backend is implemented using Python and FastAPI. It retrieves earthquake observations from the USGS GeoJSON feed, validates and normalizes the information, and stores it in PostgreSQL using SQLAlchemy. The backend also provides APIs for earthquake alerts, context-area analysis and route suggestions. Mapbox is used to request route alternatives when valid destination and configuration data are available.

Database

SafeMyanmar uses PostgreSQL as its backend database for normalized earthquake records. The mobile application uses Drift for offline caching and Flutter Secure Storage for local profile and emergency-contact information. This design allows important information to remain available when network connectivity is limited.

Earthquake Data Processing

Earthquake information is collected from trusted APIs and processed through cloud and edge computing. The system analyzes magnitude, location, and severity, then generates alerts and safety recommendations. HPC resources can be used to run simulations and predict aftershock risks, improving preparedness.

Navigation System

The navigation system relies on GPS and edge computing to recommend safe evacuation routes. It considers blocked roads, flooded areas, fire zones, and nearby shelters. Routes are continuously updated as disaster conditions change, ensuring context-aware guidance rather than static directions.

SOS System

The SOS feature allows users to quickly send their GPS location, emergency message, and timestamp to family members, contacts, and (in future versions) rescue teams. The Rescue Beacon Mode complements this by activating alarms, flashing the flashlight in SOS patterns, and displaying a HELP message to aid victim identification.

Offline Emergency Guide

Recognizing that connectivity may fail during disasters, SafeMyanmar includes an offline emergency guide. This knowledge base covers CPR, bleeding control, burns, fractures, and disaster-specific safety measures such as earthquake and flood survival tips. It ensures that essential instructions remain accessible without internet access.

AI Assistance

The emergency assistant primarily uses deterministic intent matching and a reviewed offline knowledge base. This allows important guidance to remain available without an internet connection. The architecture also supports optional on-device AI models, but critical medical, SOS and route-related guidance remains controlled and reviewed.

Testing and Evaluation

Testing Approach

SafeMyanmar was tested using a combination of **functional testing**, **user evaluation**. The goal was to ensure that each feature worked correctly under disaster-like conditions and that the system remained reliable even with limited connectivity.

Testing focused on:

- Verifying **core features** such as alerts, navigation, SOS, and AI guidance.
- Ensuring **usability** for diverse user groups (citizens, families, rescue teams).

Functional Testing

Each major feature was tested to confirm correct operation:

- **Disaster Alerts:** Verified that alerts were received from trusted sources and displayed promptly.
- **Safe Route Navigation:** Tested GPS-based routing under simulated blocked roads and hazards.
- **SOS System:** Confirmed that location and emergency messages were sent to contacts.
- **Rescue Beacon Mode:** SafeMyanmar supports nearby SOS communication through Bluetooth Low Energy. After explicit user confirmation, the application can broadcast limited emergency information to nearby SafeMyanmar devices. The information may include a temporary event identifier, timestamp, approximate location status and battery level. Sensitive profile, contact and message information is not included in the Bluetooth broadcast.
- **AI Assistant:** Evaluated both online (cloud-based) and offline (knowledge base) responses.
- **Offline Emergency Guide:** Ensured accessibility of first aid instructions without internet.

User Evaluation

SafeMyanmar includes automated tests for mobile interfaces, location states, SOS preparation, Bluetooth communication, offline caching, earthquake-data processing, API endpoints, navigation services, database operations and security-related behaviour.

- Earthquake information was displayed using available test data.
- The main application screens and navigation operated correctly.
- SOS messages could be prepared after explicit confirmation.
- Nearby SOS communication was tested using Bluetooth functionality.
- Emergency guidance remained accessible offline.
- English and Myanmar interfaces displayed correctly.

Limitations and Future Improvements

Limitations

SafeMyanmar also has several limitations.

- Disaster information depends on the availability and reliability of external data sources.
- GPS may not always be accurate, especially in difficult environments.
- Safe-route recommendations depend on available road, hazard, and shelter information.
- AI-generated assistance may require careful validation to avoid incorrect emergency advice.
- Cloud-based features depend on Internet connectivity.
- SOS communication may be affected by network availability.
- Rescue Beacon location sharing may not work when there is no network connection.
- Some features, such as damage reporting and advanced disaster analysis, are future enhancements.

Future Improvements

Future versions of SafeMyanmar could include:

- Integration with more official Myanmar disaster-warning systems
- Real-time disaster notifications
- More accurate evacuation-route analysis
- Wider shelter coverage
- Disaster damage reporting
- AI-based disaster image recognition
- Damage-severity prediction
- Personalized emergency recommendations
- Advanced earthquake simulations

- Flood prediction
- Large-scale evacuation optimization
- Integration with rescue organizations
- Expanded Bluetooth and device-to-device communication

Conclusion

SafeMyanmar demonstrates how modern computing technologies can be integrated into a single mobile application to provide **trusted disaster alerts, safe navigation, emergency communication, rescue support, AI guidance, and offline first aid resources**. By combining mobile computing, ubiquitous computing, GPS, edge computing, cloud computing, AI, and HPC, the system offers faster, more reliable, and context-aware disaster assistance.

Although limitations remain, the proposed future improvements highlight the potential for SafeMyanmar to evolve into a comprehensive national disaster-response platform. Its ultimate goal is to reduce disaster response time, save lives, and support both citizens and organizations during emergencies.

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Appendices

No additional appendices are included because all relevant project information is already presented in the main sections of this report.